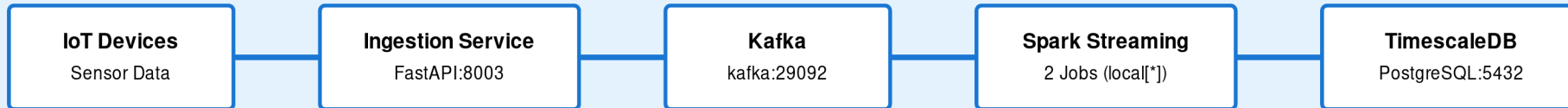
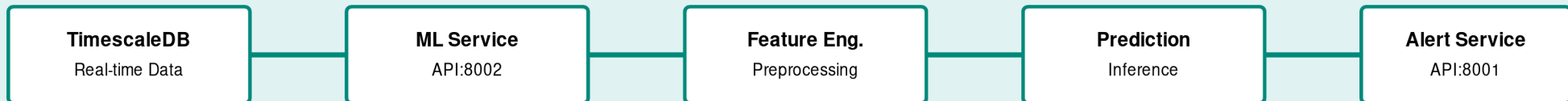


Flow 1: Data Ingestion Pipeline



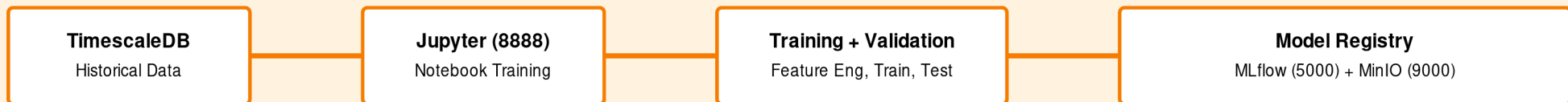
Purpose: Real-time sensor data ingestion from industrial equipment to time-series database
Characteristics: High throughput | Low latency | Fault tolerance | Exactly-once semantics

Flow 2: ML Inference Pipeline



Purpose: Real-time anomaly detection using trained machine learning models
Characteristics: <50ms inference | XGBoost/RandomForest models | Alert Detector worker | Kafka alerts topic

Flow 3: Model Training Pipeline



Purpose: Periodic model training using historical sensor data for anomaly detection
Characteristics: Batch processing | Experiment tracking | Model versioning | Performance validation